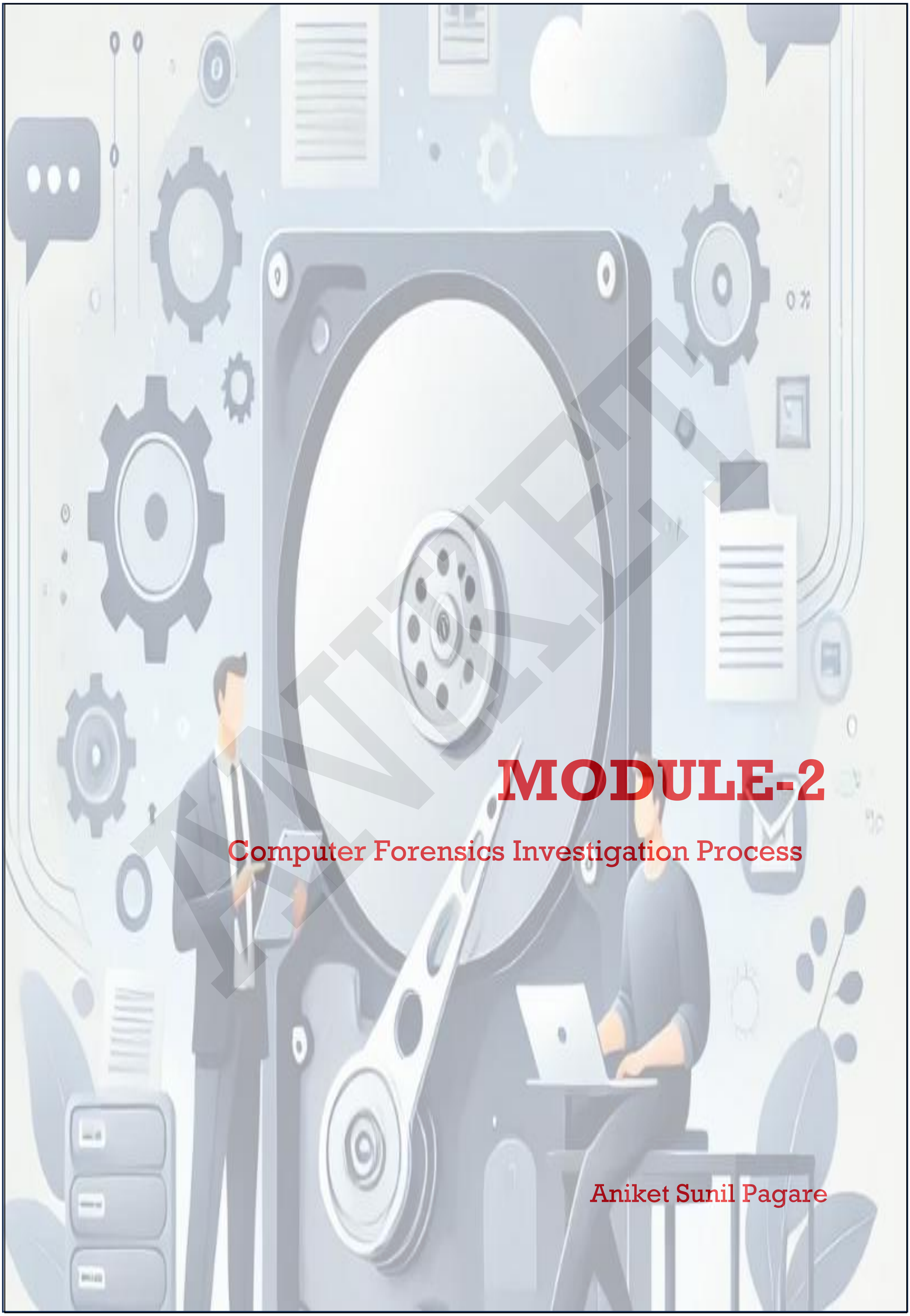


MODULE-2

Computer Forensics Investigation Process

Aniket Sunil Pagare

Computer Forensics Investigation Process

The **Computer Forensics Investigation Process** is a structured approach to **collecting, preserving, analyzing, and presenting digital evidence** in a legally acceptable way. It ensures the evidence remains **authentic, reliable**, and **admissible in court**.

📚 Phases of Computer Forensics Investigation

🔵 1. Identification

"What happened, where, and on which devices?"

Purpose: Recognize and define the scope of the incident.

Tasks:

- Determine the type of incident (fraud, data theft, hacking, etc.).
- Identify the devices involved (PCs, servers, phones, USBs).
- Decide what data might be relevant (emails, logs, files).

🟢 2. Preservation

"Secure and protect the data from being altered or lost."

Purpose: Ensure integrity and prevent tampering.

Tasks:

- **Isolate** affected systems (disconnect from the network).
- **Create bit-by-bit forensic images** (not just file copies).
- **Document everything** – who, when, how.
- Hash the data using **MD5/SHA-1/SHA-256** to prove integrity.

🟠 3. Collection

"Collect evidence in a legally sound and forensically valid way."

Purpose: Gather all possible digital evidence.

Tasks:

- Acquire physical devices (hard drives, USBs, phones).
- Dump memory (RAM) for volatile data.
- Copy logs from firewalls, routers, mail servers.
- Extract emails, browser history, registry entries, deleted files.

🔴 4. Examination

"Organize and search for relevant evidence."

Purpose: Analyze collected data for patterns and hidden information.

Tasks:

- Recover deleted files.
- Examine file timestamps (MAC times).
- Extract chat logs, documents, emails, images.
- Detect signs of anti-forensics (data hiding, wiping).

Techniques:

- Keyword search
- Metadata analysis
- File carving
- Steganography detection

🟣 5. Analysis

"What does the evidence say?"

Purpose: Make sense of the evidence and link it to the incident.

Tasks:

- Timeline reconstruction (who did what, and when).
- Identify attacker's methods and goals.
- Find user activity (logins, browsing, downloads).

- Compare against known signatures (e.g., malware indicators).

⚫ 6. Documentation

"Create a detailed report with findings and evidence."

Purpose: Record every action for legal and internal purposes.

Tasks:

- Write **step-by-step logs** of how evidence was collected, analyzed, and what was found.
- Include screenshots, hash values, timelines, file metadata, etc.
- Maintain **chain of custody logs**.

⚪ 7. Presentation

"Present your findings in court or to an internal authority."

Purpose: Deliver findings in a clear, accurate, and non-technical way for stakeholders.

Tasks:

- Explain technical concepts in plain language.
- Answer questions from judges, lawyers, or company officials.
- Be prepared to testify as an expert witness.

Data Recovery

Data recovery is the process of **retrieving lost, deleted, corrupted, damaged, or inaccessible data** from storage devices such as **hard drives, SSDs, USB drives, memory cards, CDs, mobile phones**, and even **cloud storage**.

Objectives-:

- **Recover accidentally deleted files** from storage devices.
- **Restore data lost due to formatting** or partition errors.
- **Retrieve files from corrupted or inaccessible systems.**
- **Recover data from physically damaged devices.**
- **Minimize data loss during system failures or crashes.**
- **Maintain business continuity by restoring critical information.**
- **Preserve digital evidence for forensic investigations.**
- **Recover backup or cloud-stored data after incidents.**
- **Access files encrypted or affected by malware or ransomware.**
- **Ensure data integrity and confidentiality during recovery.**

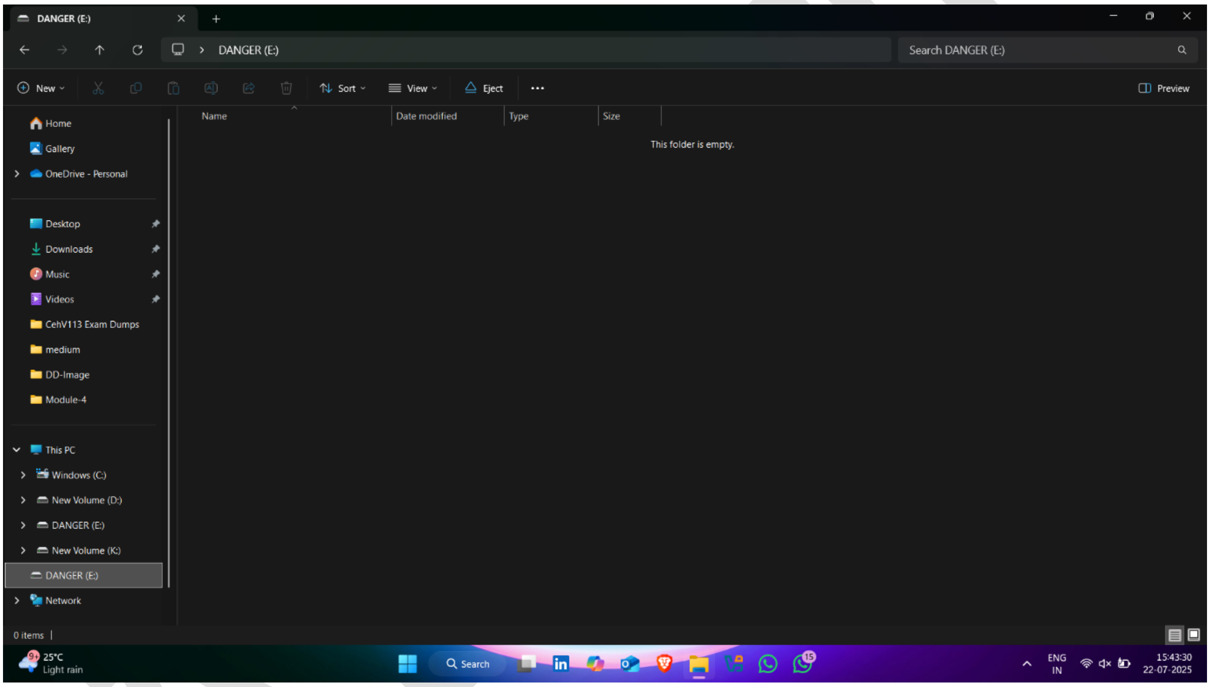
A) Data Recovery Using Recuva

Recuva is a **free data recovery software** developed by **Piriform (the makers of CCleaner)**. It helps users **recover accidentally deleted files** from Windows computers, USB drives, memory cards, external hard drives, and other storage devices.

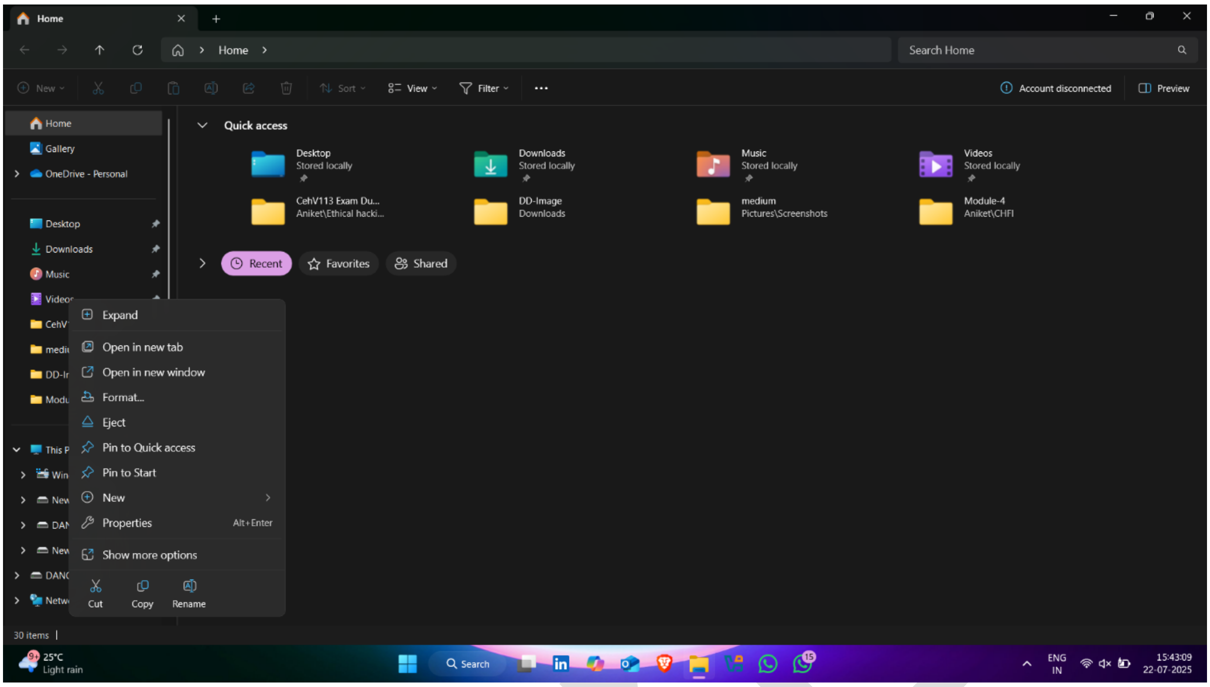
Download Link : [rcsetup154]()

How to use it -:

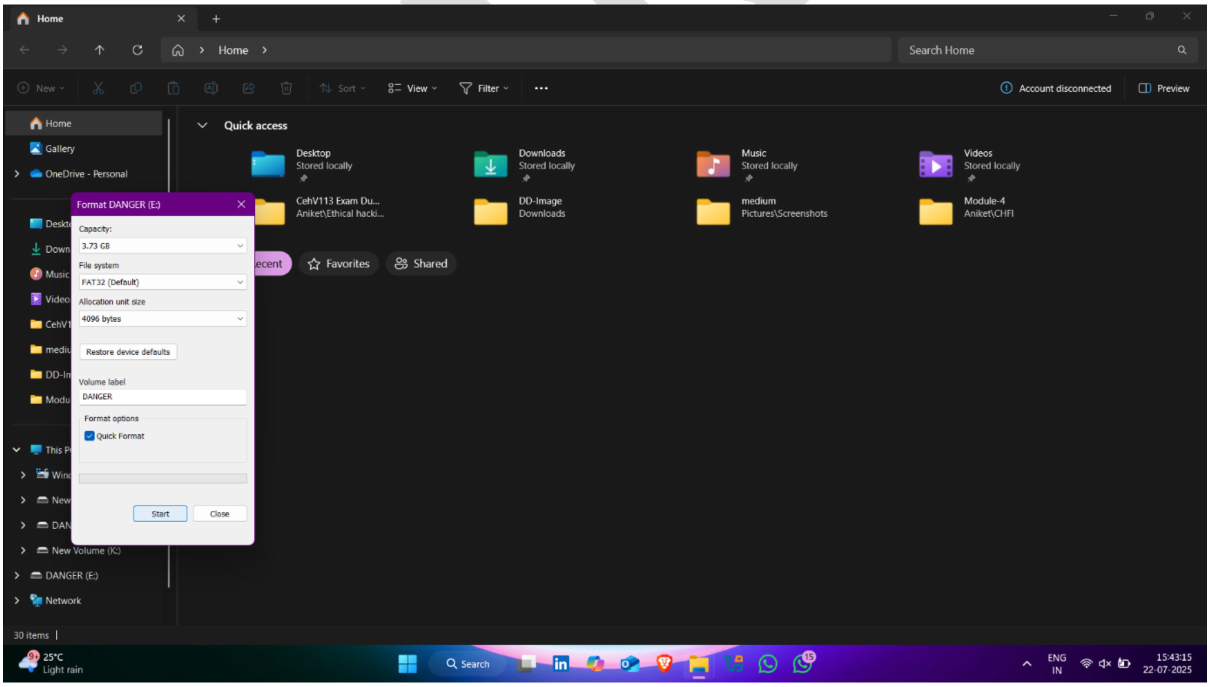
- **I have Usb Pendrive and want to recover this pendrive**
- Usb Pendrive Device Empty 👇 ✅



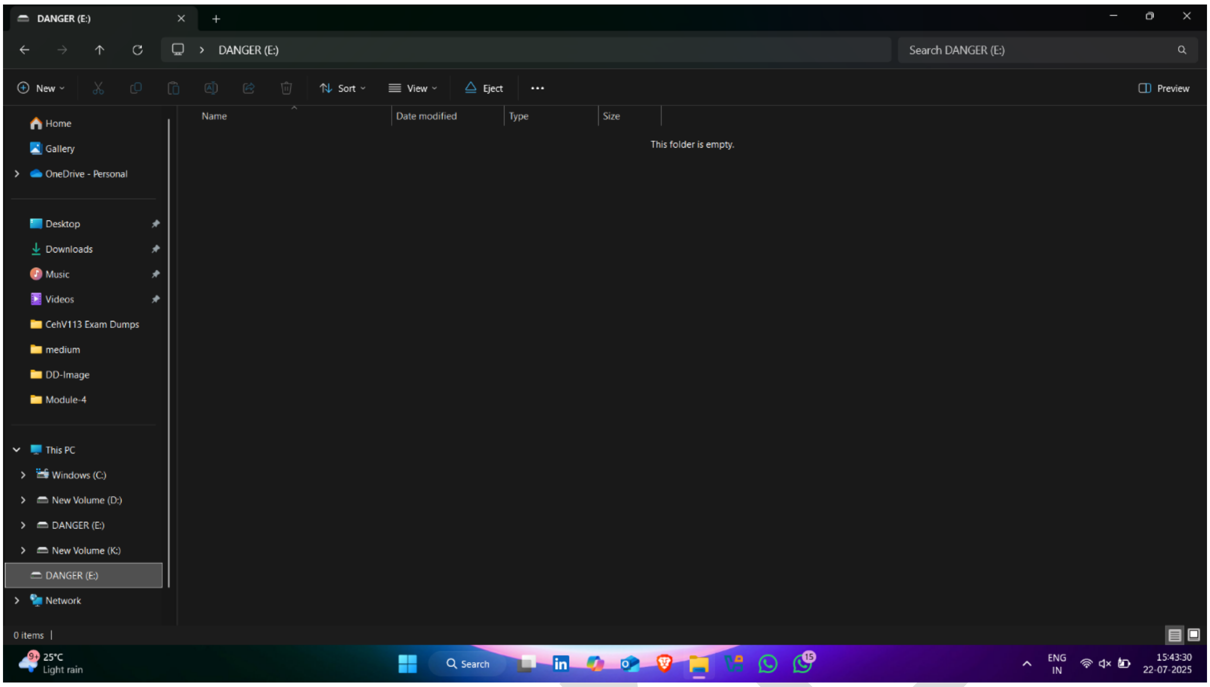
- Right click on device and click on format option



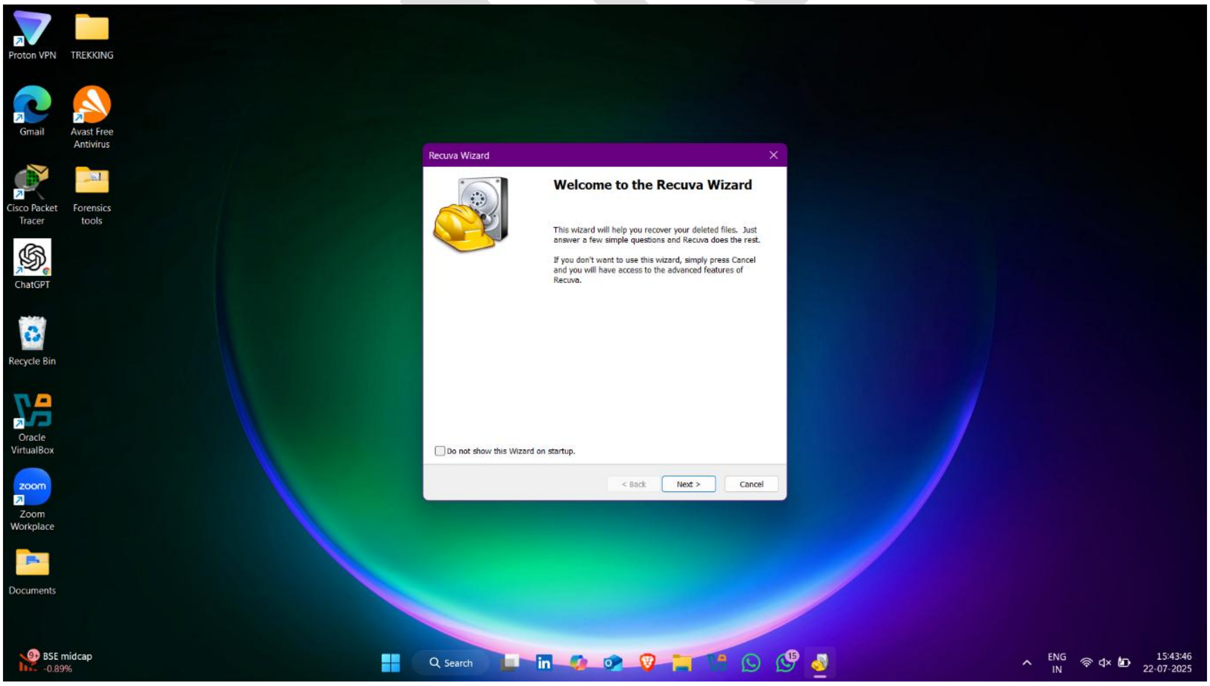
- Click on start



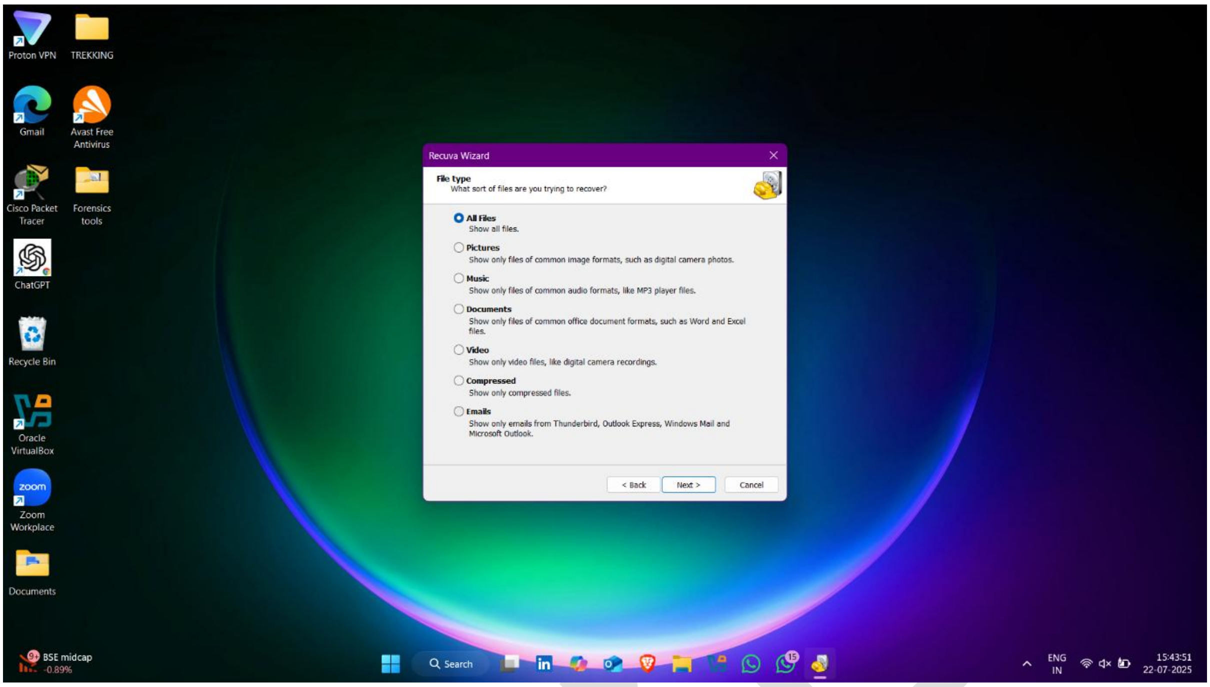
- Pendrive formatted



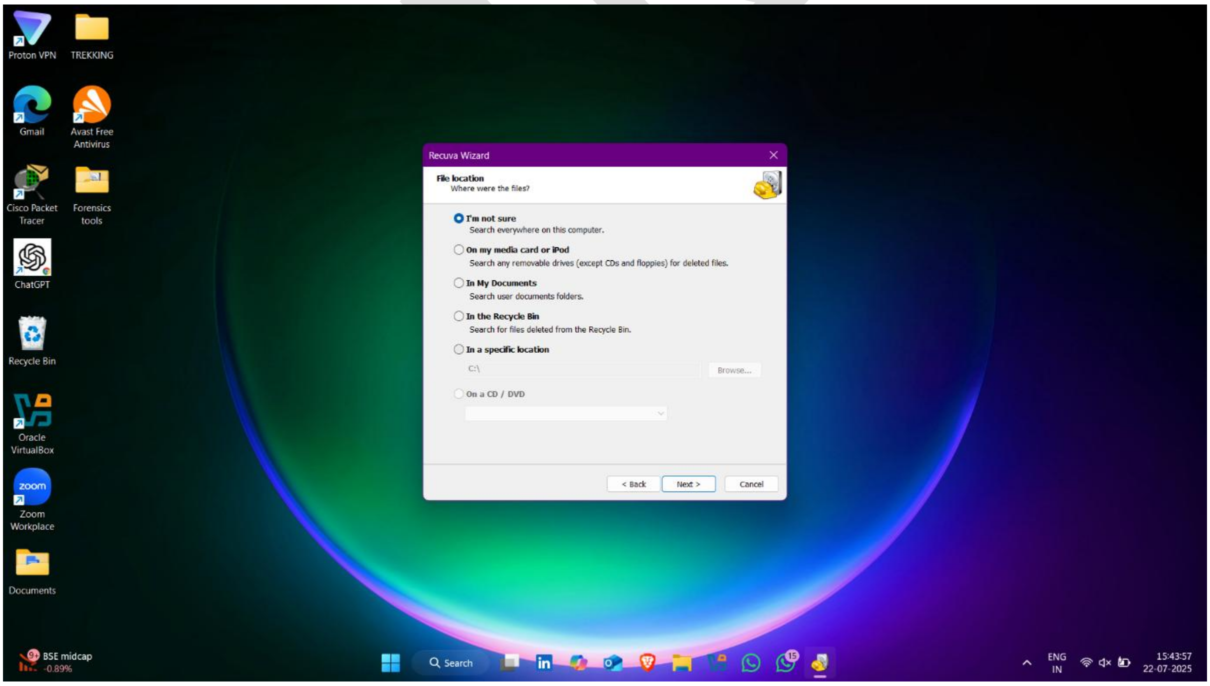
- Now open recuva tool



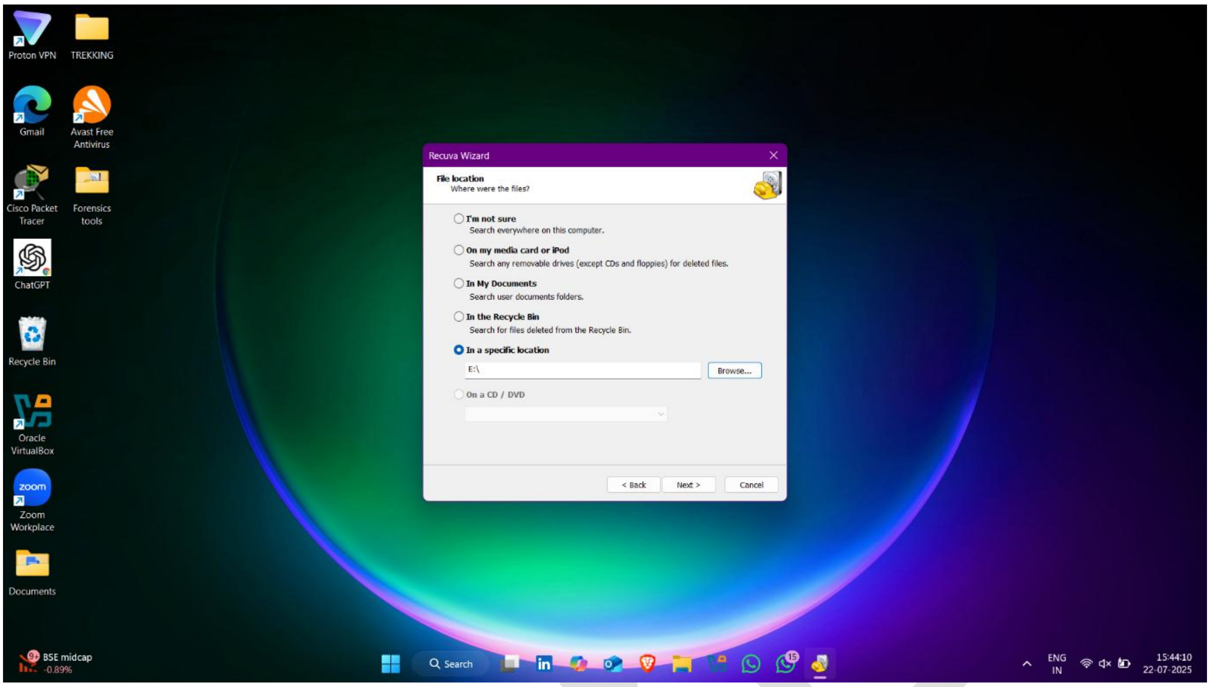
- Select **all files** option and click on **next**



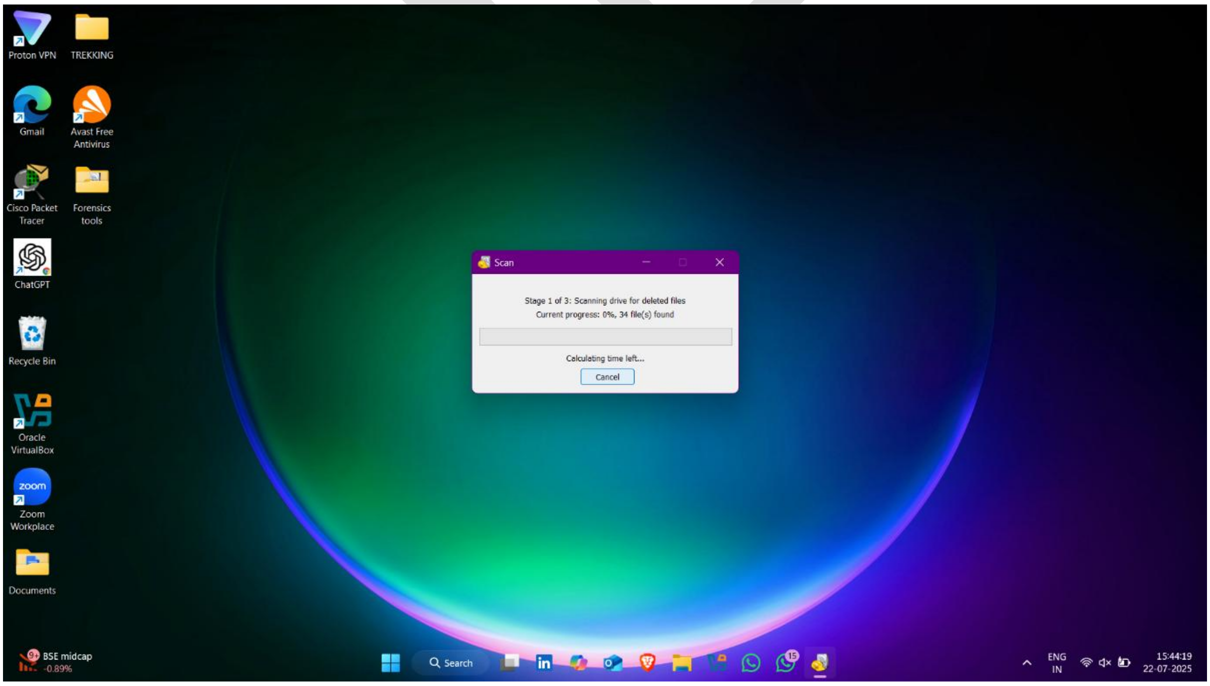
- Click on **in the specific location**



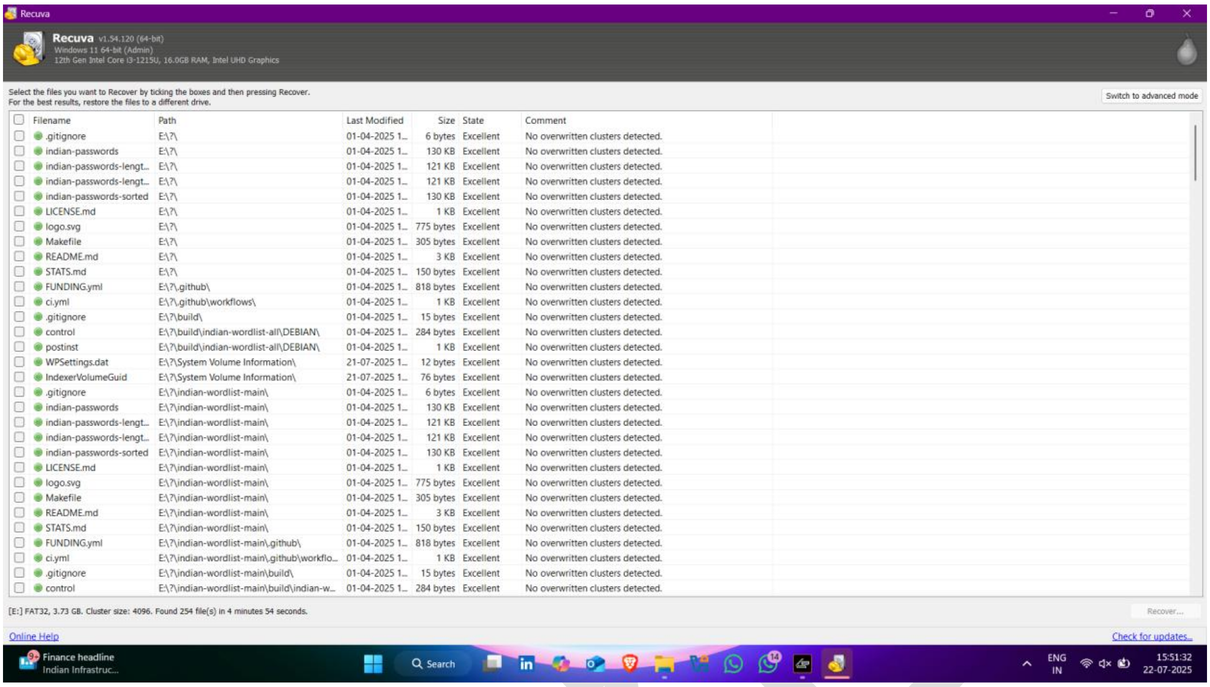
- Click on **Browse** option and select the **usb pendrive**



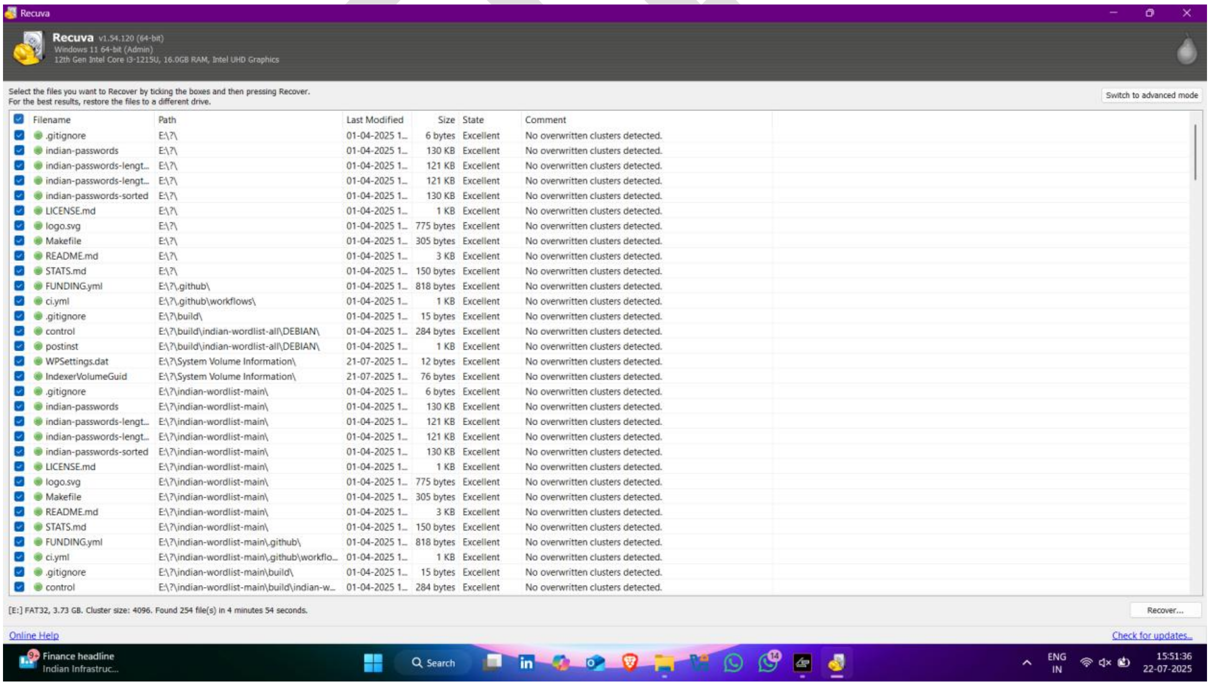
- Here recovery started ✅ 👇



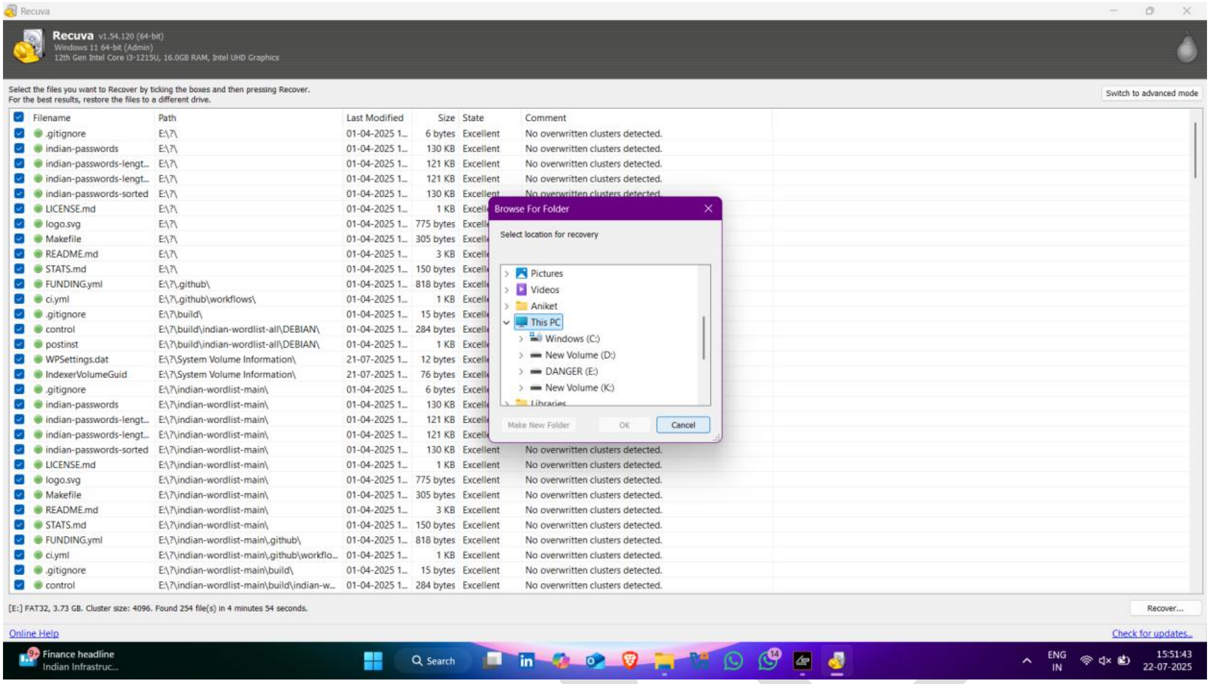
- Here data recover ✅ 👇



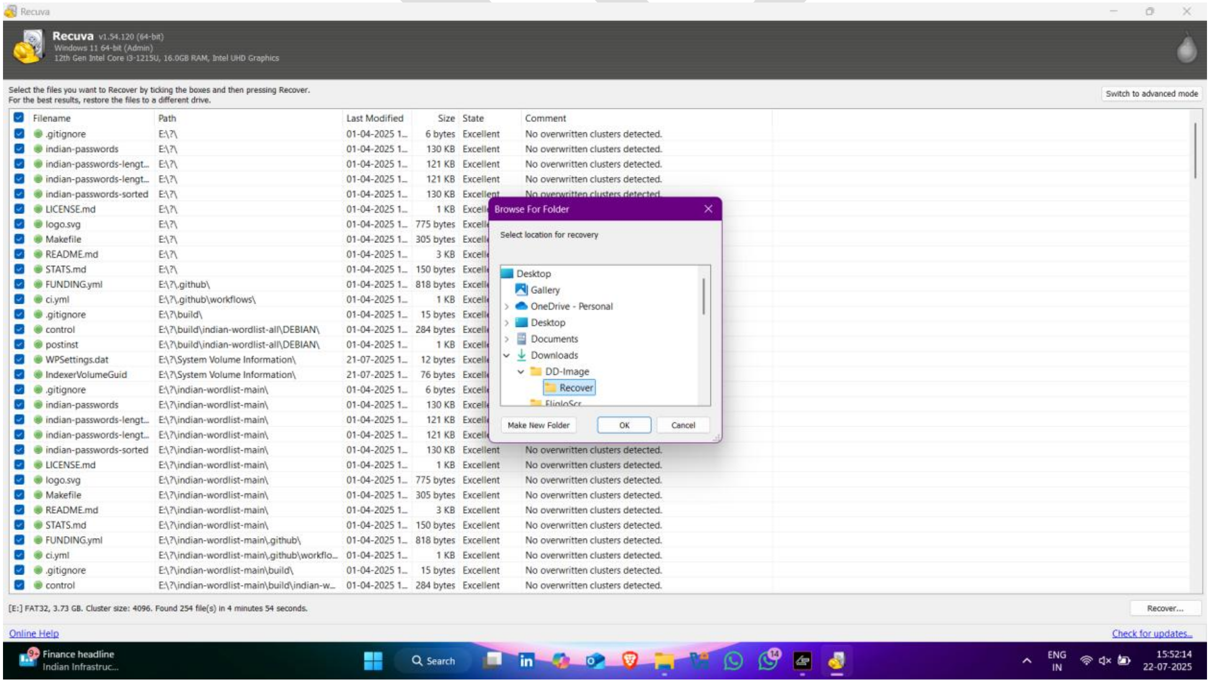
- Now **select all** the data and click on **Recover option (right side bottom)**



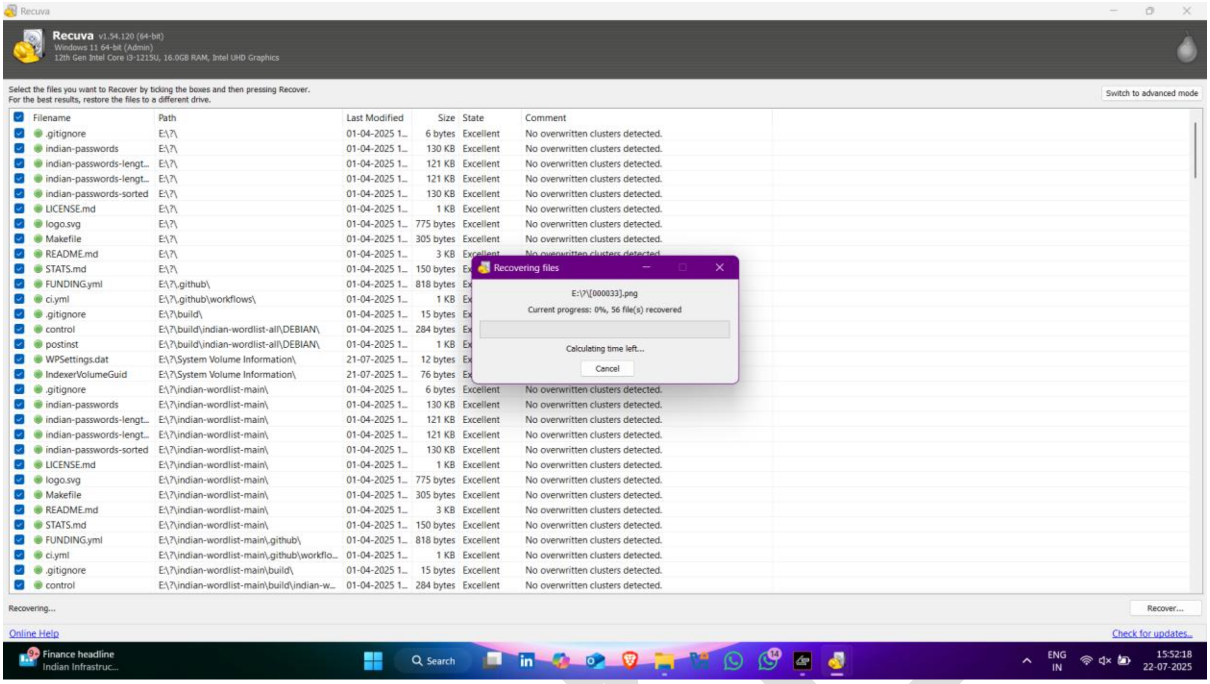
- Now select the location that you want to stored the recover files



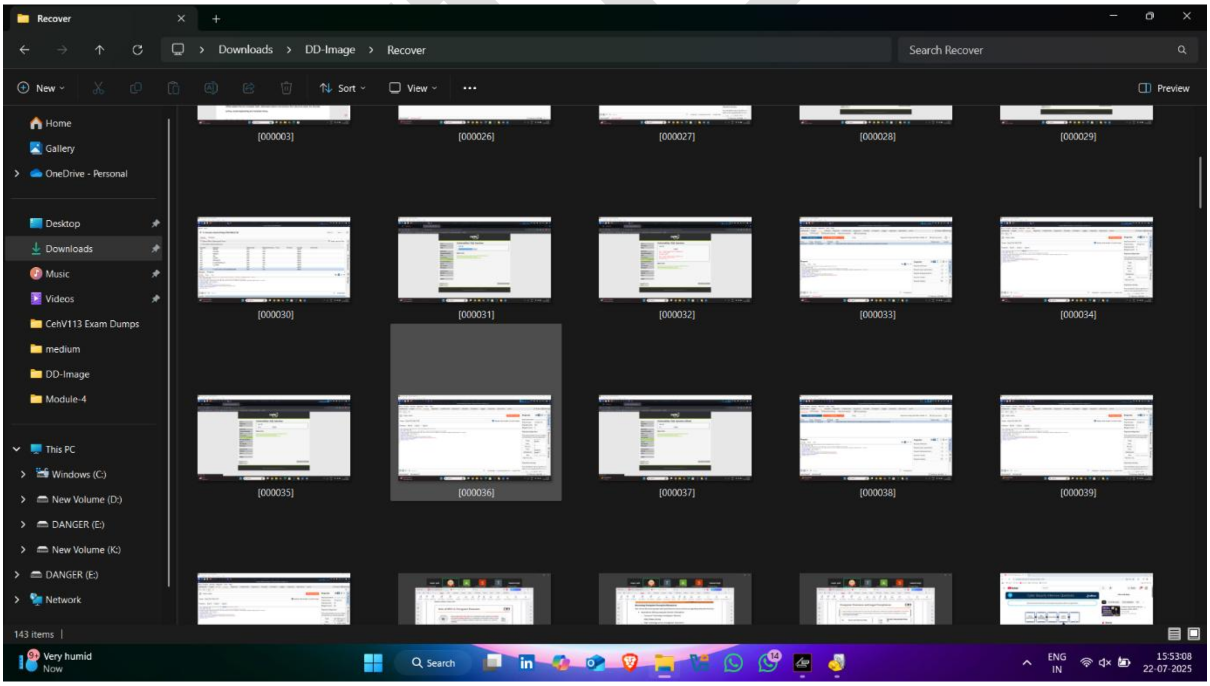
- Click on ok



- Storing Recover Data Started ✅ 👇



- Recover Data ✅ 👇



B) Data Recovery Using EaseUP Tool

EaseUS Data Recovery Wizard is a **powerful, user-friendly** data recovery software that helps recover **lost, deleted, formatted, or inaccessible data** from **PCs, laptops, USB drives, SSDs, memory cards, and external hard drives**.

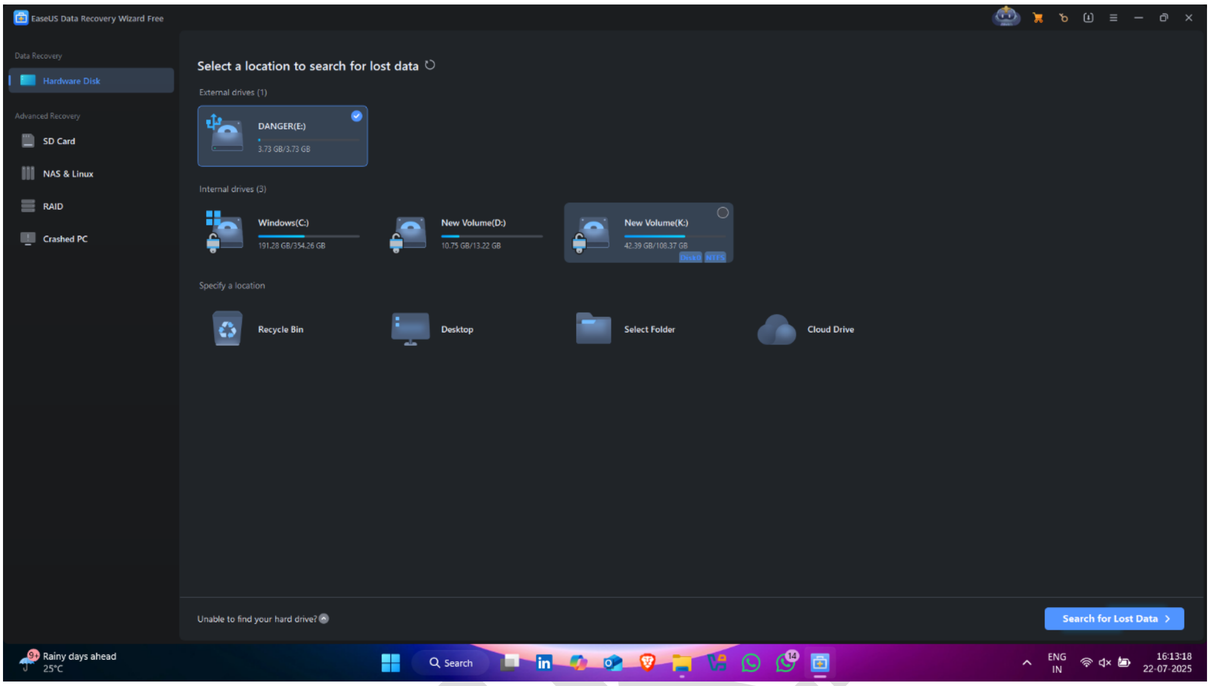
🎯 **Objectives of EaseUS Tool :**

1. **Recover deleted files** from Recycle Bin, hard drives, or partitions.
2. **Restore formatted data** from internal or external drives.
3. **Retrieve lost partitions** and their data due to partition errors.
4. **Recover data from crashed or unbootable systems.**
5. **Extract files from RAW or inaccessible drives.**
6. **Support recovery after virus or malware attacks.**
7. **Recover a wide range of file types** (photos, videos, documents, emails, etc.).
8. **Offer preview before recovery** to ensure correct file selection.
9. **Support both Windows and macOS platforms.**
10. **Provide professional-grade recovery with minimal user effort.**

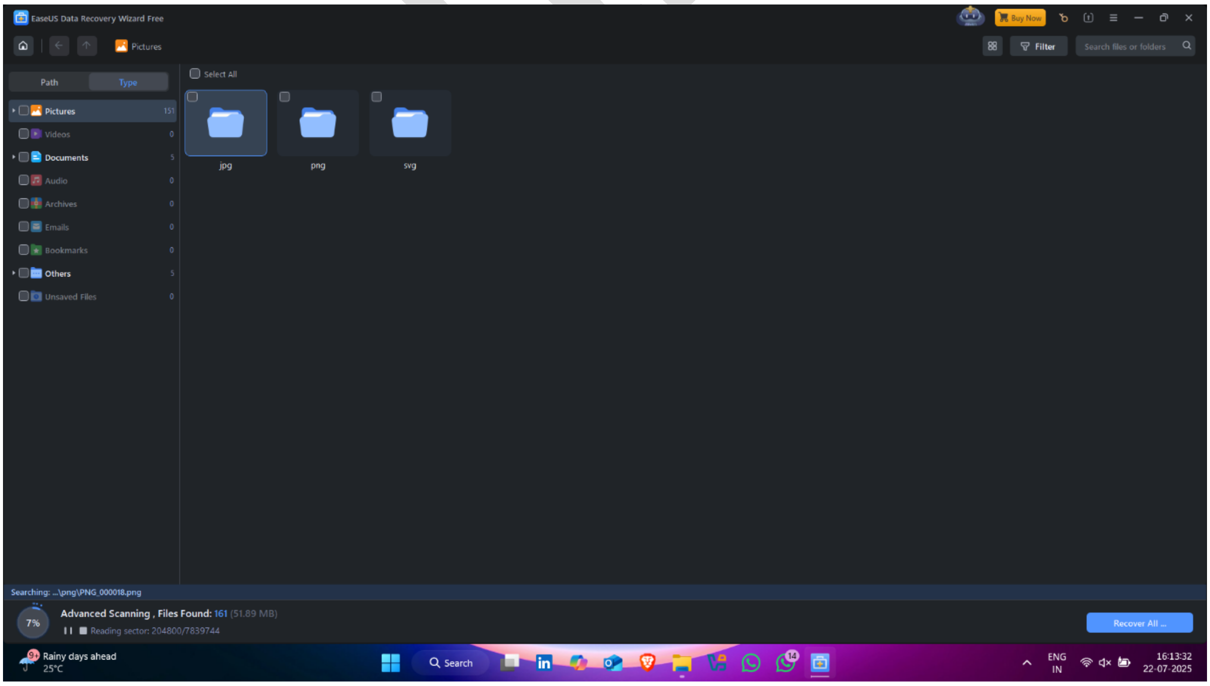
Download Link :
https://www.easeus.com/datarecoverywizardpro/

How to use it -:

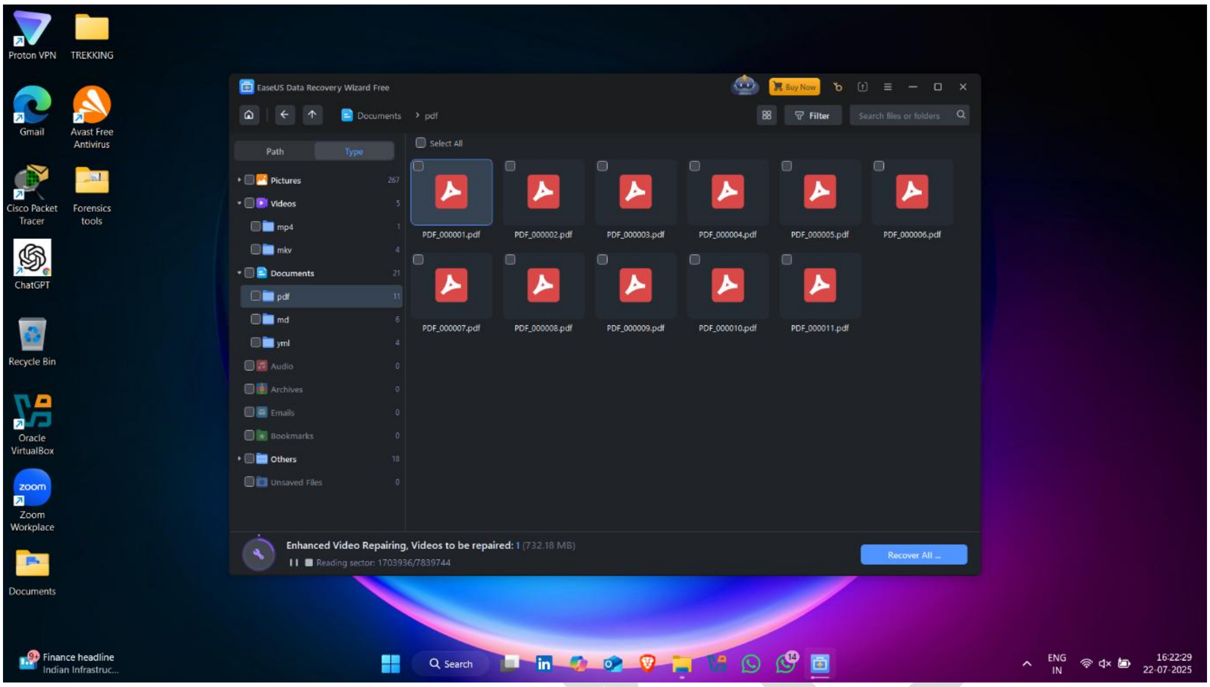
- Click on drive that you want to recover data
- Click on **Usb Pendrive**



- Recovery Started 👇 ✅



- **Here data recovered ✅ 👇**



Handle Evidence Data

Handling evidence data refers to the proper collection, preservation, documentation, and protection of digital evidence to maintain its integrity and admissibility in legal proceedings.

🎯 Objectives of Handling Evidence Data (One-liners):

1. ✅ **Maintain evidence integrity** — Ensure data is not altered or tampered with.
2. 🧾 **Preserve chain of custody** — Document every access or transfer of evidence.
3. 🛡️ **Ensure legal admissibility** — Follow legal procedures for court acceptance.
4. 🔒 **Prevent data loss or corruption** — Use forensically sound methods/tools.
5. 🕵️ **Enable repeatable analysis** — Allow independent verification of findings.
6. 📁 **Secure storage** — Store evidence in protected and access-controlled environments.
7. 👨‍⚖️ **Support investigation and prosecution** — Provide reliable data to legal authorities.

A) Create Image Using FTK Imager

FTK Imager (Forensic Toolkit Imager) is a **free digital forensic acquisition tool** developed by **AccessData**. It is used to **create forensic images (copies) of data** from hard drives, USBs, or other storage media **without altering the original evidence**.

🎯 Purpose of FTK Imager:

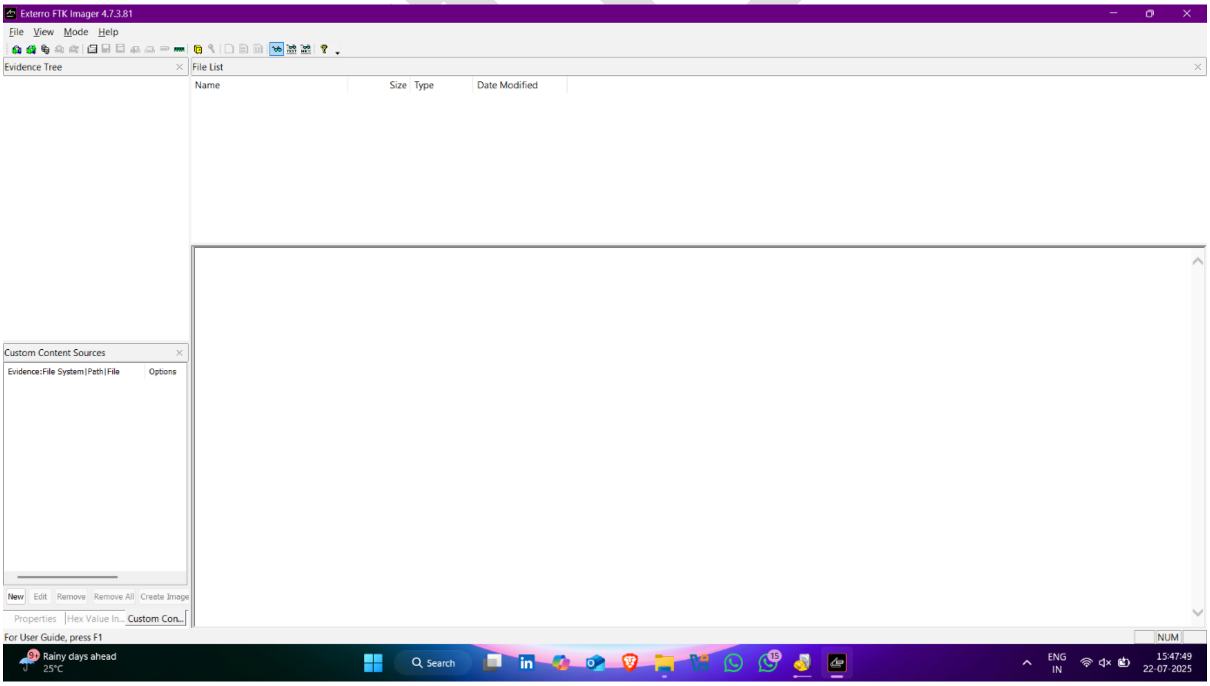
1. 🧪 **Create bit-by-bit forensic images** of storage media.

2. 🔍 **Preview files and folders** before imaging (even deleted files).
3. 🧾 **Verify integrity** of data using **hash values** (MD5, SHA1, SHA256).
4. 📦 **Export specific files/folders** from evidence.
5. 🔐 **View protected and hidden files** in a readable format.
6. 💾 **Create E01, AFF, or RAW image formats** for analysis in forensic tools.
7. 🛡️ **Ensure evidence preservation** for legal use.

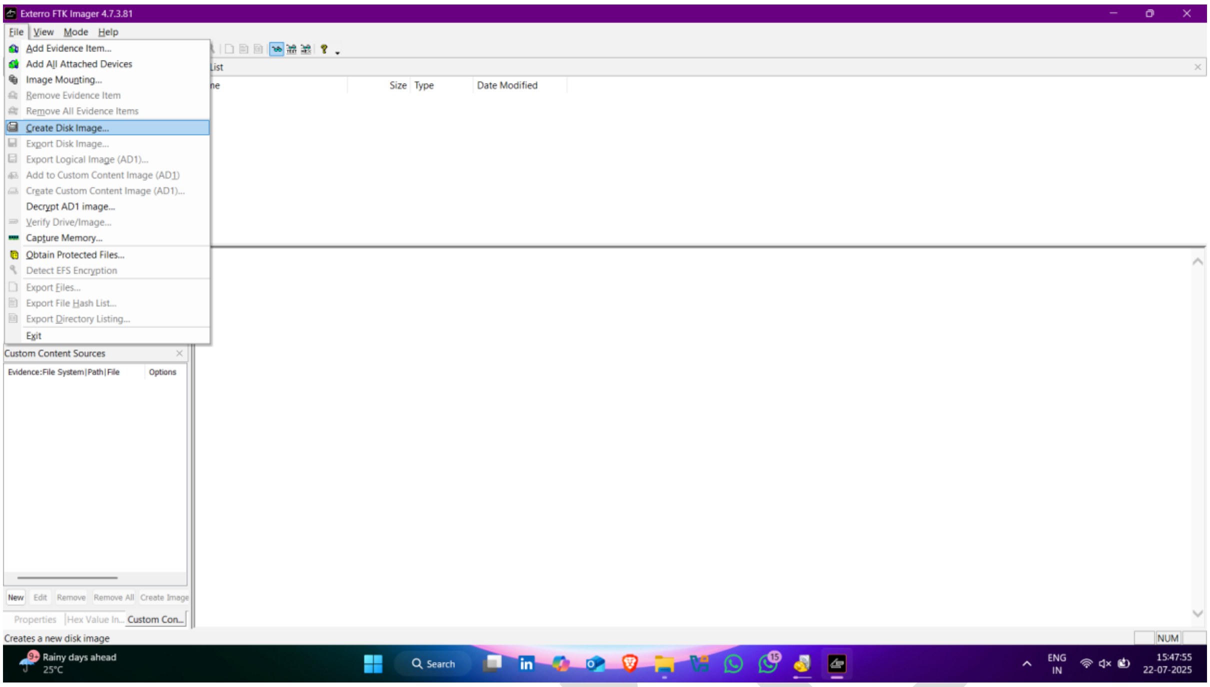
Download Link : https://www.exterro.com/ftk-product-downloads/ftk-imager-4-7-3-81

How to use it -:

- Open FTK Imager 👇 **,** and click on **File** Option



- Click on **Create Disk Image**



- Pop up appears ✅ 👇

🔍 **Select Source - Evidence Type Options:**

1. 📀 **Physical Drive**
   - This captures an **entire hard drive**, including deleted files, unallocated space, and slack space.
   - 🔧 Used when full forensic imaging is needed.

---

2. 💾 **Logical Drive**
   - This captures only **active partitions** (like C:, D:), not deleted files or unallocated space.
   - ⚠️ Lighter analysis, used when full image isn't needed.

---

3. 🖼️ **Image File**
   - Load an existing forensic image file (like .E01, .dd, or .img) for analysis.

- 📂 Used to analyze previously acquired images.

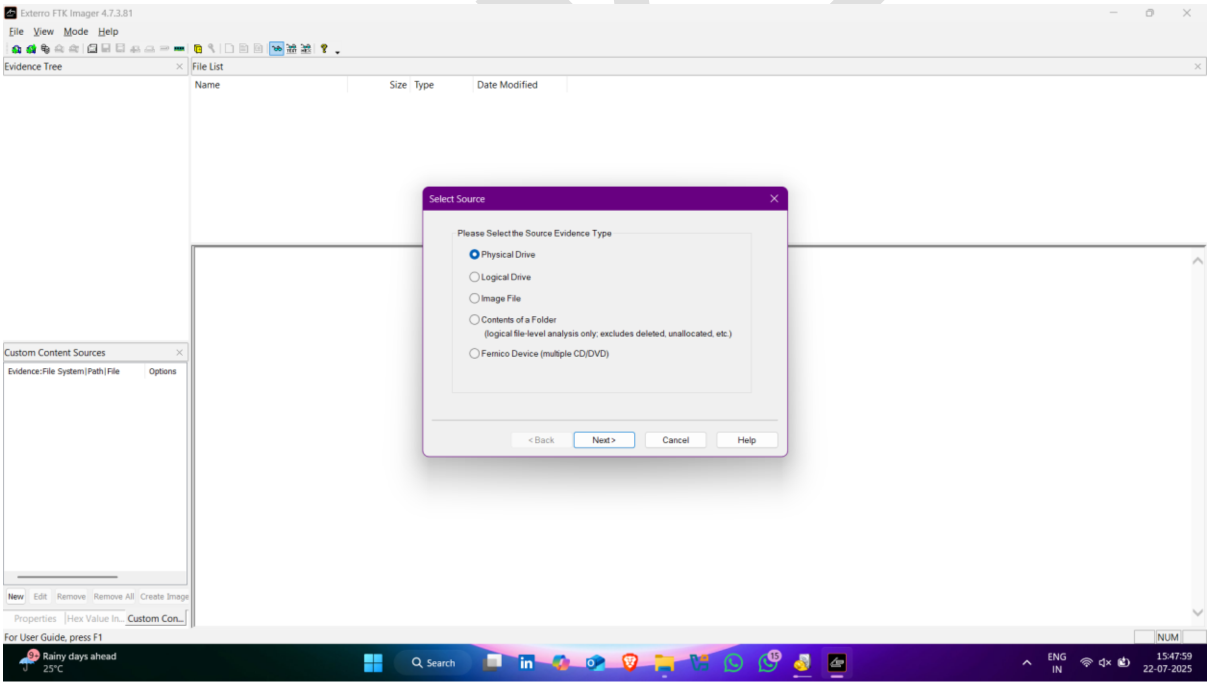
---

4. 📁 **Contents of a Folder**
   - Acquires and analyzes **only files/folders** from a specified directory.
   - ❌ Does **not include deleted or hidden data**.

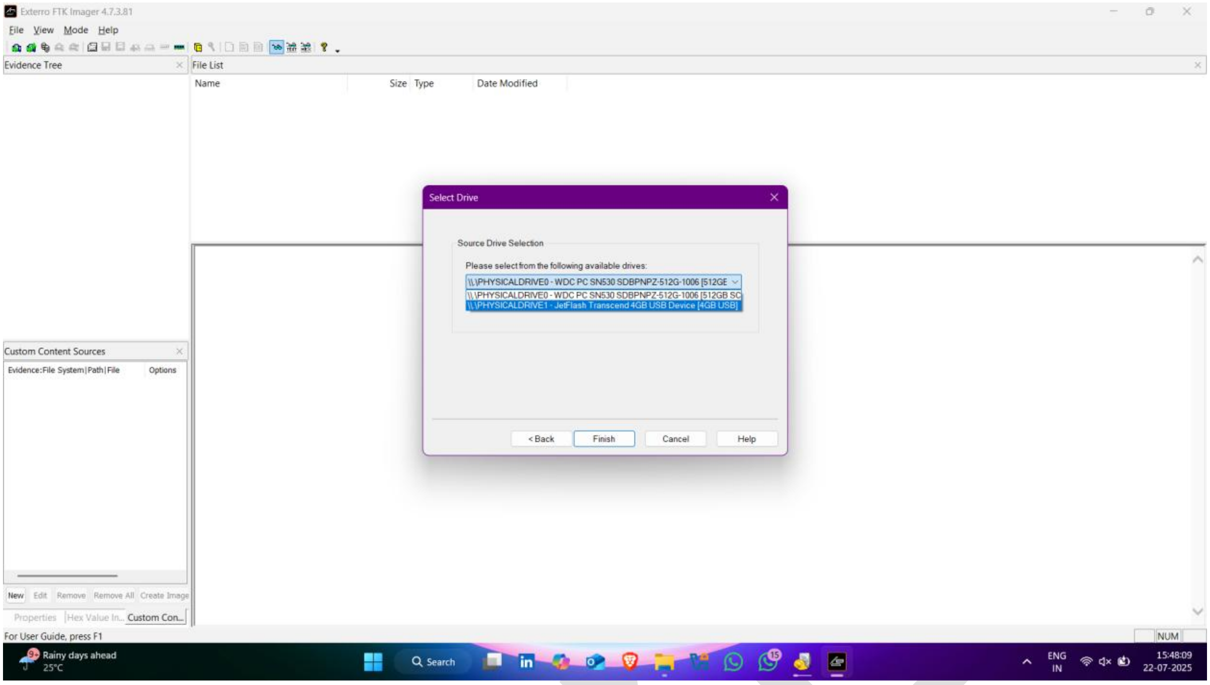
---

5. 💽 **Fermico Device (CD/DVD)**
   - Acquires data from **optical media** like CDs or DVDs.
   - 📀 Useful for examining data stored on physical media.

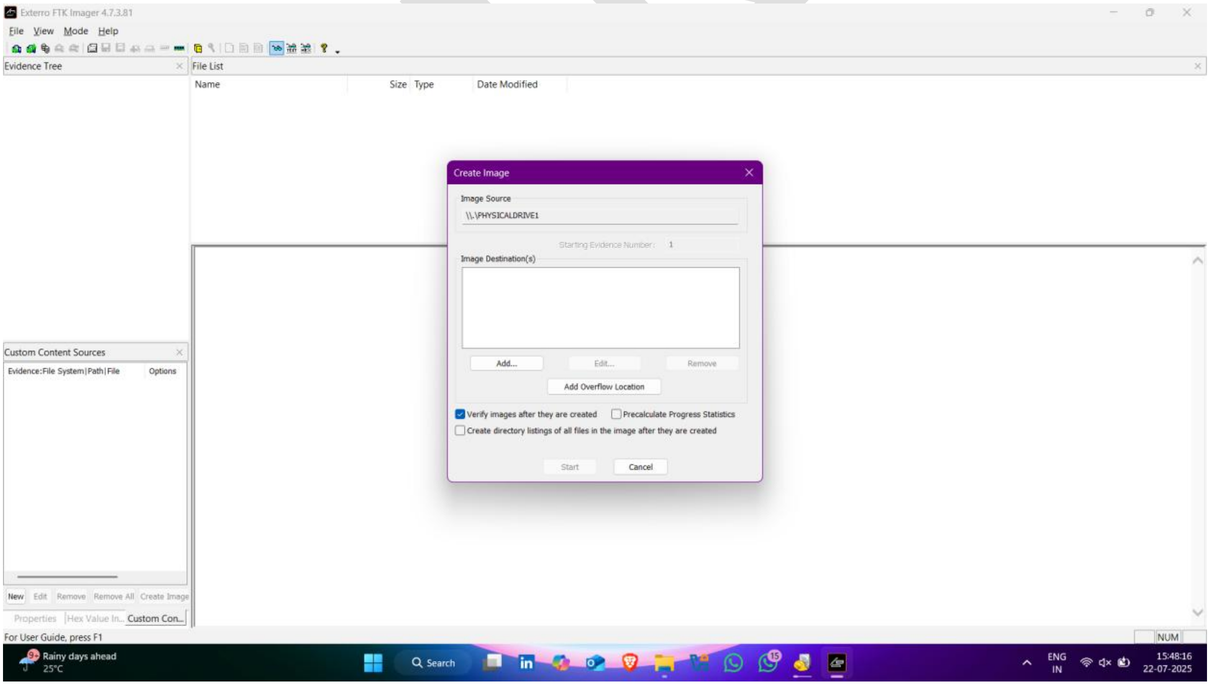
- Click on **First Option** and then click on **next**



- Select **pendrive** and click on **Finish**



- Click on **add** to set destination



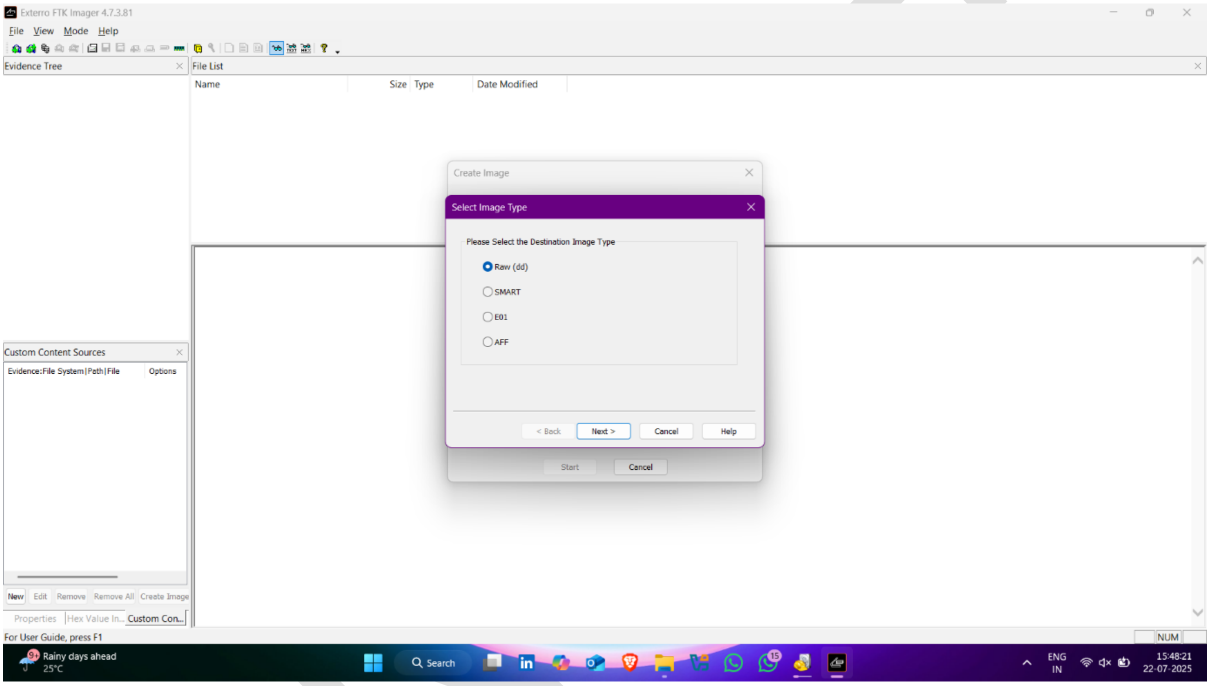
- Now select the format of image file 👇

1. 🔵 **Raw (dd)** – *(Default selected)*
    - Creates a **bit-by-bit copy** of the source.
    - Extension: .dd or .img
    - ✅ Pros: Universally supported by most forensic tools.
    - ❌ Cons: No compression or metadata.
2. 📘 **SMART**
    - Proprietary format used by **AccessData SMART**.
    - Extension: .s01
    - ✅ Pros: Supports segmented image files.
    - ❌ Cons: Less widely supported.
3. 🟡 **E01 (Expert Witness Format)**
    - Most commonly used forensic format.
    - ✅ Pros:
        - Supports compression.
        - Can include metadata (case number, examiner info, hash, etc.)
        - Supports splitting into segments.
    - ❌ Cons: Slightly slower to create and read than Raw.
    - 📁 Extension: .e01, .e02, etc. (if split)
4. 🟢 **AFF (Advanced Forensics Format)**
    - Open-source format developed for forensics.
    - ✅ Pros: Compression + metadata.
    - ❌ Cons: Less common in some tools.

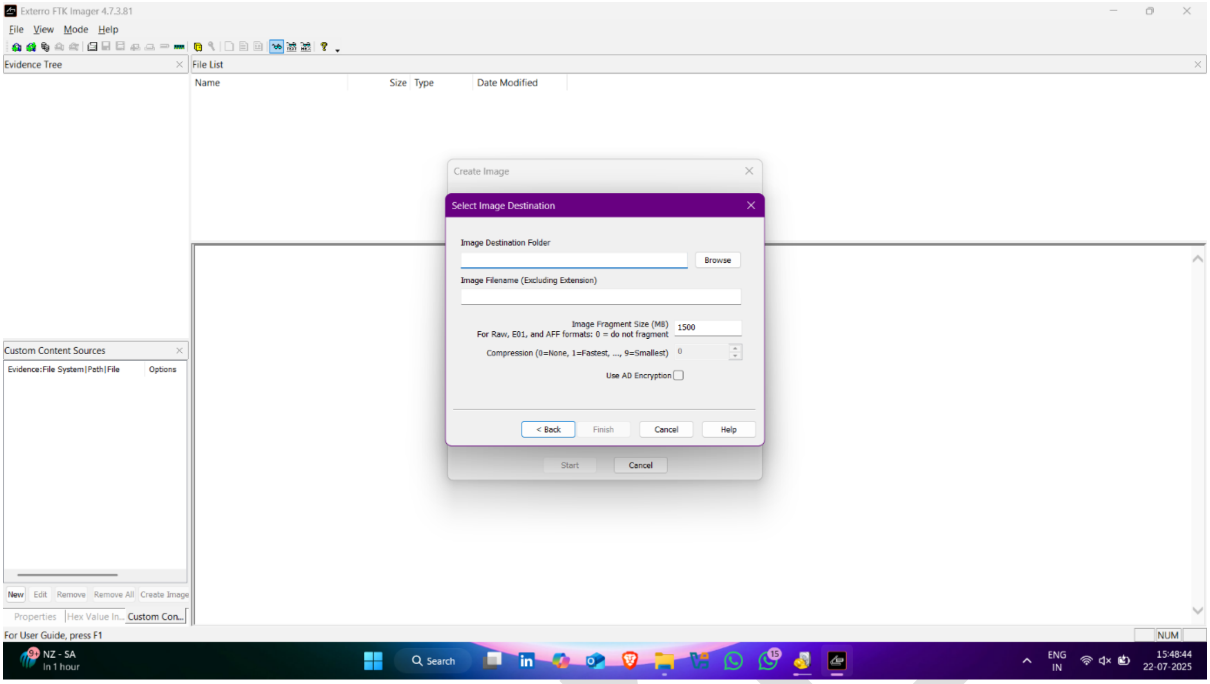
🧠 Which Should You Choose?

- ✅ **Use Raw (dd)** if you want a **simple bit-for-bit** copy and maximum compatibility.
- ✅ **Use E01** if you want compression, case metadata, and hashing included (recommended for forensic investigations).

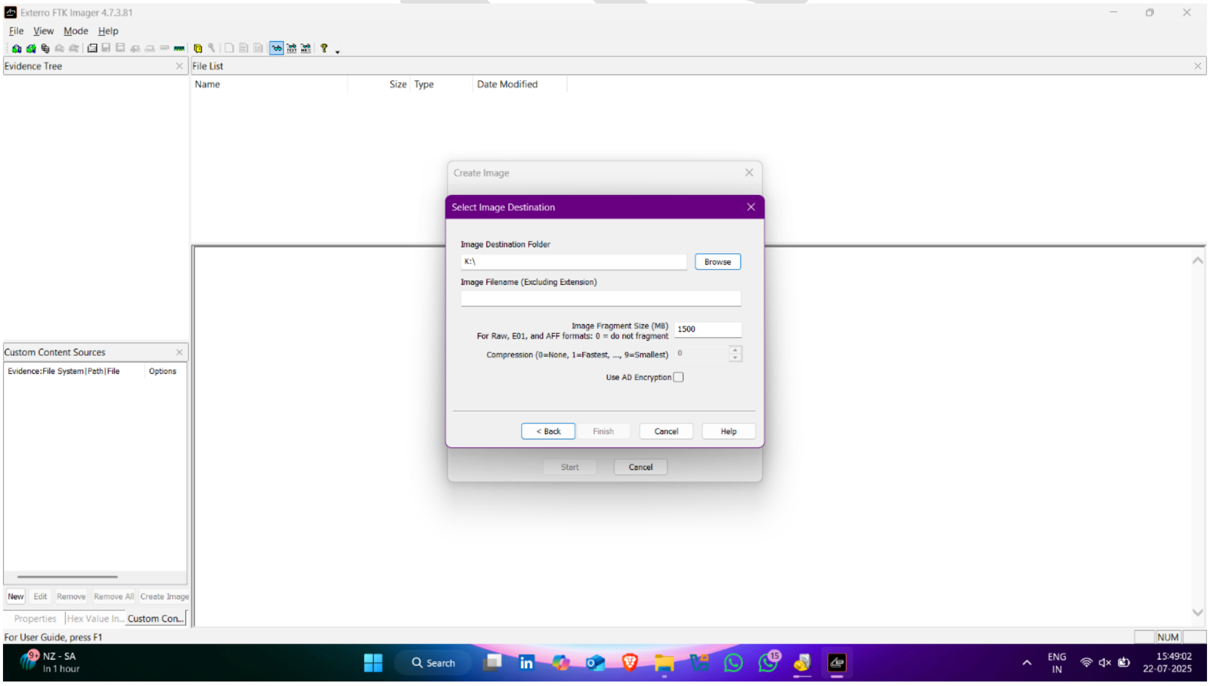
- Click on next



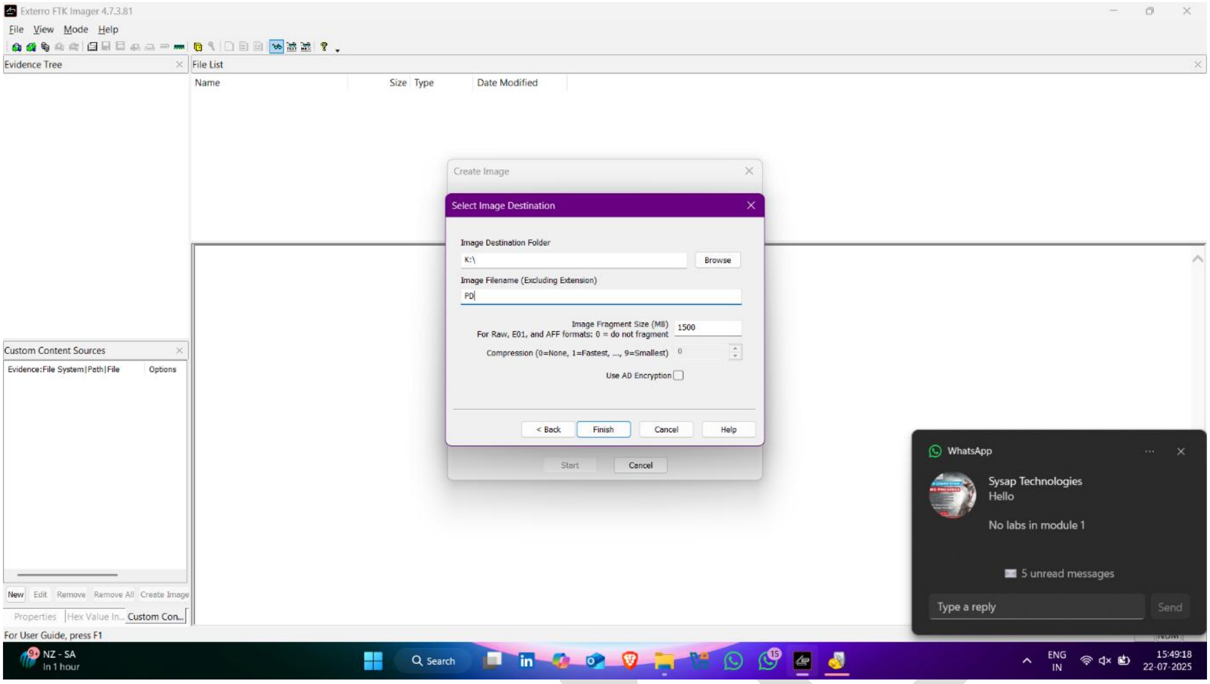
- Click on **browse** to select Destination of image file



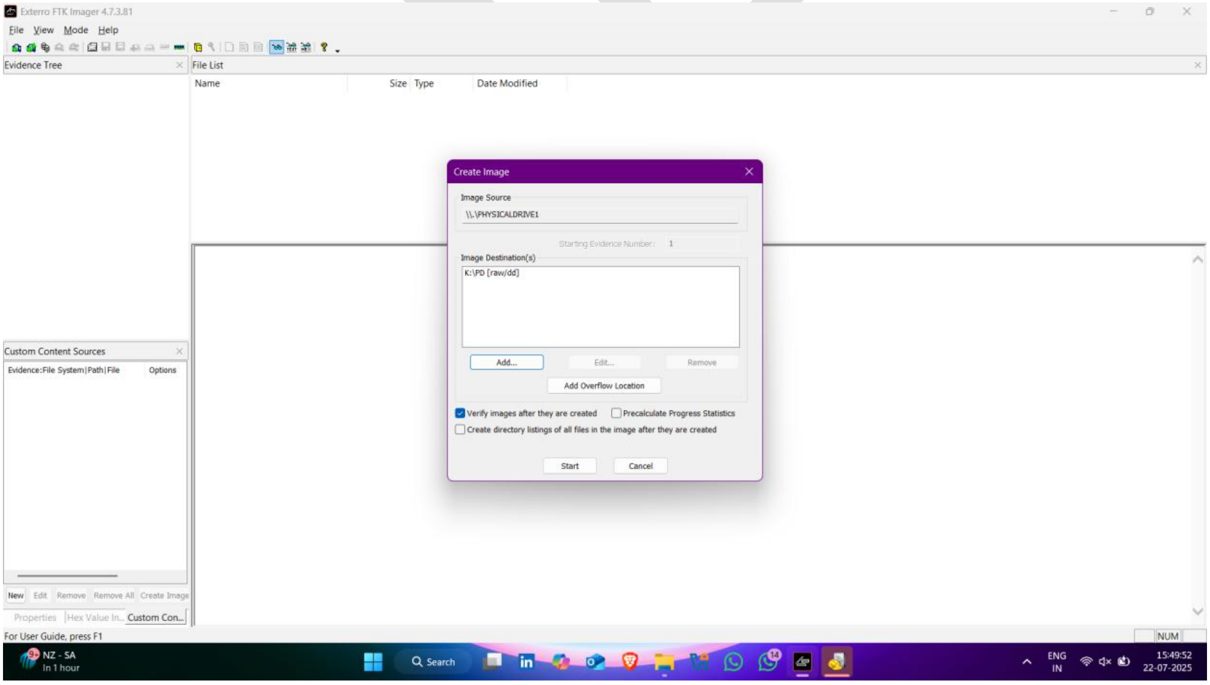
- Now enter image file name



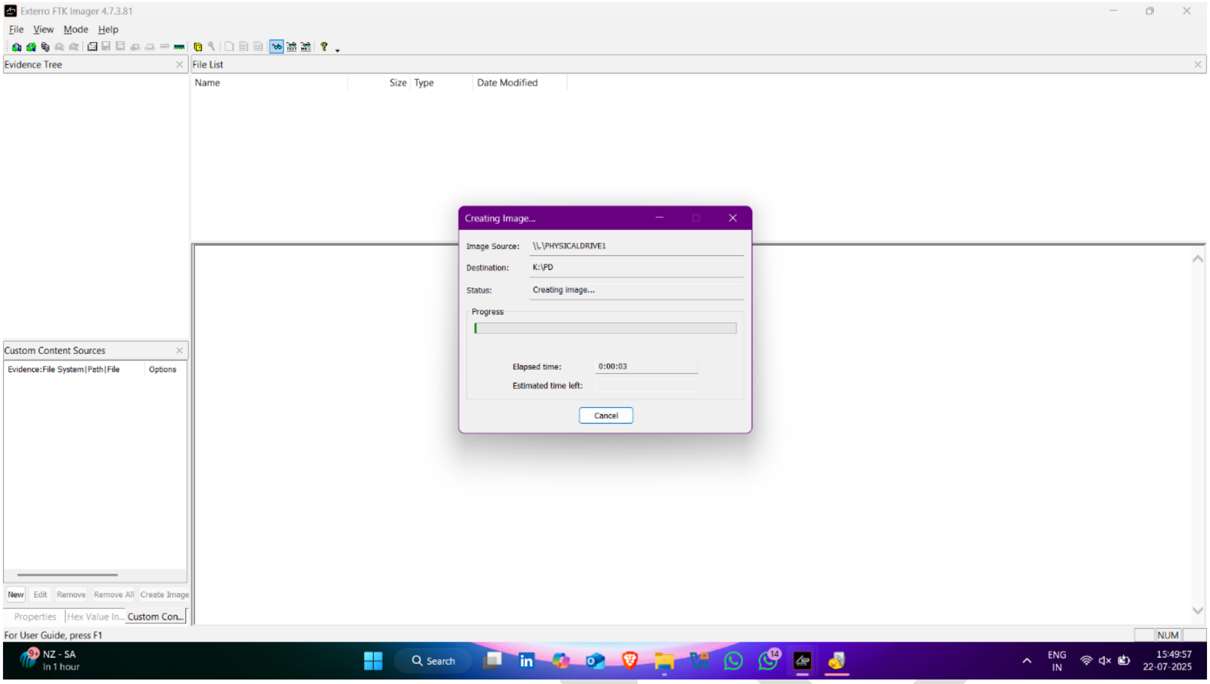
- Click on **Finish**



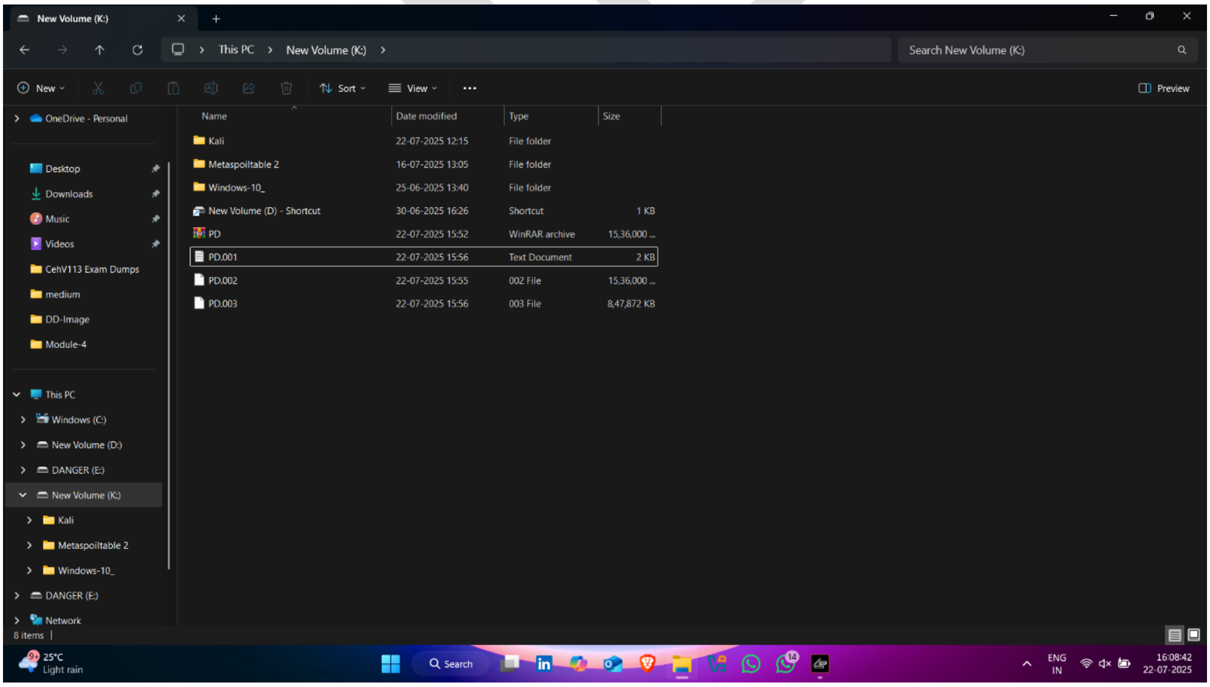
- Now click on **start** ✅



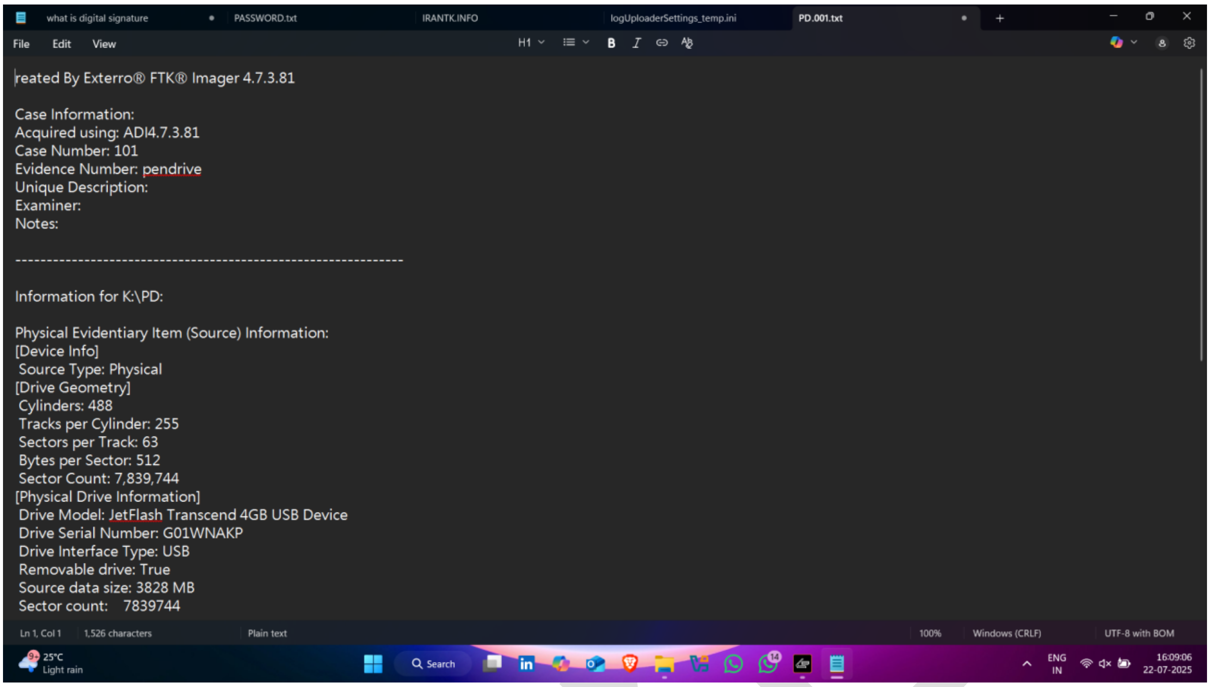
- Image creation started



- Here pendrive image created ✅ 👇

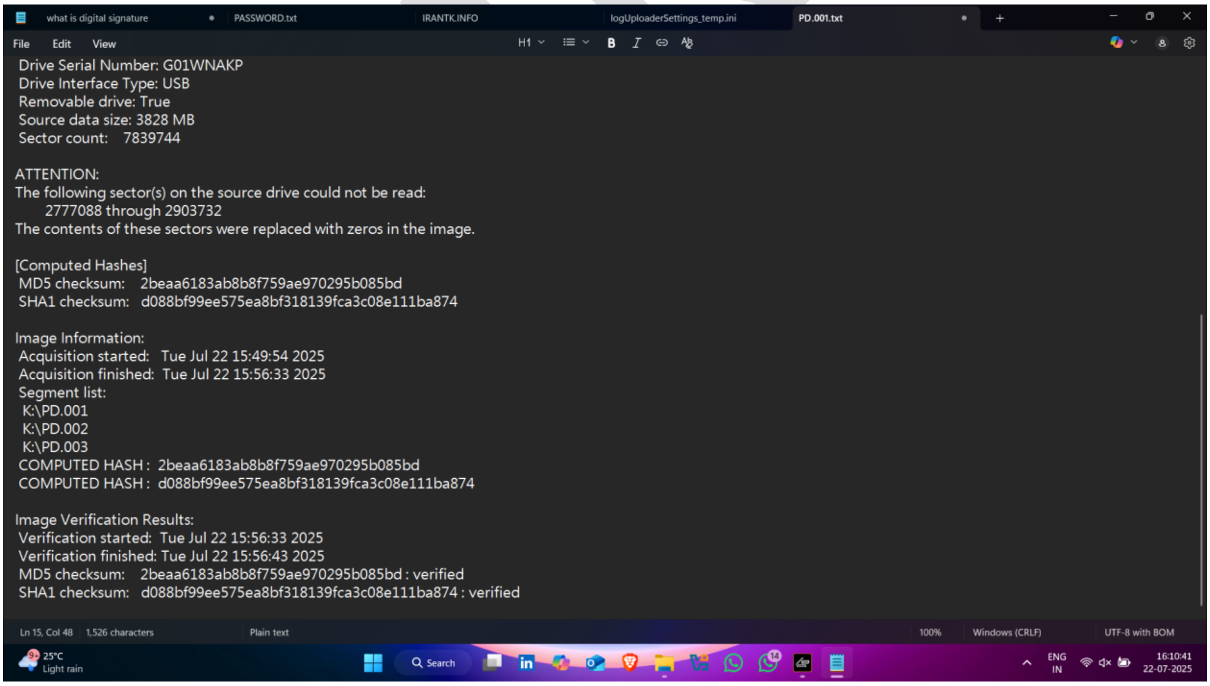


- Detailed about image



```
reated By Exterro® FTK® Imager 4.7.3.81

Case Information:
Acquired using: ADI4.7.3.81
Case Number: 101
Evidence Number: pendrive
Unique Description:
Examiner:
Notes:

--------------------------------------------------------------

Information for K:\PD:

Physical Evidentiary Item (Source) Information:
[Device Info]
 Source Type: Physical
[Drive Geometry]
 Cylinders: 488
 Tracks per Cylinder: 255
 Sectors per Track: 63
 Bytes per Sector: 512
 Sector Count: 7,839,744
[Physical Drive Information]
 Drive Model: JetFlash Transcend 4GB USB Device
 Drive Serial Number: G01WNAKP
 Drive Interface Type: USB
 Removable drive: True
 Source data size: 3828 MB
 Sector count:    7839744
```

- Different Hash values of images



```
 Drive Serial Number: G01WNAKP
 Drive Interface Type: USB
 Removable drive: True
 Source data size: 3828 MB
 Sector count:    7839744

ATTENTION:
The following sector(s) on the source drive could not be read:
	2777088 through 2903732
The contents of these sectors were replaced with zeros in the image.

[Computed Hashes]
 MD5 checksum:    2beaa6183ab8b8f759ae970295b085bd
 SHA1 checksum:   d088bf99ee575ea8bf318139fca3c08e111ba874

Image Information:
 Acquisition started:  Tue Jul 22 15:49:54 2025
 Acquisition finished:  Tue Jul 22 15:56:33 2025
 Segment list:
  K:\PD.001
  K:\PD.002
  K:\PD.003
 COMPUTED HASH : 2beaa6183ab8b8f759ae970295b085bd
 COMPUTED HASH : d088bf99ee575ea8bf318139fca3c08e111ba874

Image Verification Results:
 Verification started:  Tue Jul 22 15:56:33 2025
 Verification finished: Tue Jul 22 15:56:43 2025
 MD5 checksum:    2beaa6183ab8b8f759ae970295b085bd : verified
 SHA1 checksum:   d088bf99ee575ea8bf318139fca3c08e111ba874 : verified
```

🔍 Why We Create an Image File Instead of Using the Original in Computer Forensics

🧠 Introduction:

In digital forensics, **preserving the integrity and authenticity of evidence** is the most critical aspect of any investigation. When a suspect device (like a hard disk, USB, or mobile phone) is seized, investigators **do not work directly on the original media**. Instead, they create a **forensic image**, which is an **exact bit-by-bit replica** of the original data.

🎯 Main Motive:

The primary reason for creating an image file is to prevent any accidental or intentional modification of the original evidence, thereby maintaining its integrity for legal and investigative purposes.

📌 Detailed Reasons:

1. ✅ Preservation of Evidence Integrity

- The original media may contain sensitive data (e.g., deleted files, timestamps, logs).
- Even opening a file or booting a system can change **metadata** (like access time).
- Creating a forensic image ensures the original is **not altered in any way**.
- This is crucial to maintain the **chain of custody** and ensure the evidence is **admissible in court**.

---

2. 🛡️ Protection Against Data Corruption or Loss

- During investigation, forensic tools may crash or cause unintentional changes.

- If you were working on the original device, that data could be **permanently lost or corrupted**.
- By using an image, you can repeat analysis multiple times **without risk**.

---

3. 🔄 Repeatability of Tests

- Multiple investigators or tools may be used to analyze the same evidence.
- With an image, the same copy can be given to different examiners for **independent validation** of results.
- This improves **credibility** and transparency in forensic findings.

---

4. 🧪 Deep-Level Analysis

- Imaging tools capture **everything**, including:
  - Deleted files
  - Slack space
  - Unallocated sectors
  - Hidden partitions
- These areas are often **not visible** in normal file browsing but can contain critical evidence.

---

5. ⚖️ Legal Admissibility and Chain of Custody

- Courts require proof that the evidence was **not altered** after collection.
- Using an image and preserving the original as a "gold copy" shows that you **followed standard forensic procedures**.
- A proper chain of custody includes hashes (MD5, SHA1) to verify that the image file matches the original bit-by-bit.

---

6. 🧩 Allows Multiple Formats for Analysis

- Forensic images can be saved in formats like:
    - **RAW (.dd)** – bit-for-bit copy
    - **E01** – EnCase format with compression and metadata
    - **AFF** – Advanced Forensics Format
- These formats allow the use of **various forensic tools** to analyze the data efficiently.
